

SCOTTSBORO 2 NE, AL, USA

WMO: 750070

Lat: 34.6941N

Lon: 85.9998W

Elev: 194

StdP: 99.02

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -6.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 34.1	(d) N/A	(e) 32.9	(f) N/A	(g) 31.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-11.6	35.6	2.5	1.8	-13.4	36.9	-14.9	38.0	-16.3	39.0	-18.1	40.4
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.3	6.9	11.3	15.7	20.2	24.4	25.5	25.3	22.4	16.0	9.6	7.1		
	DBStd	8.55	5.80	5.51	5.03	4.09	3.44	2.02	1.94	2.09	3.19	4.62	4.58	5.23		
	HDD10.0	554	195	116	47	6	0	0	0	0	0	6	61	123		
	HDD18.3	1823	437	322	223	98	24	0	0	0	7	101	264	349		
	CDD10.0	2654	17	29	86	177	317	431	479	475	371	192	49	33		
	CDD18.3	883	0	1	3	18	83	181	221	217	127	29	2	1		
	CDH23.3	7667	0	6	56	218	749	1611	1853	1832	1062	269	11	1		
	CDH26.7	3015	0	0	6	28	219	681	776	816	421	67	2	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	1486	143	132	167	119	118	110	114	88	123	78	120	145		
	PrecMax	1956	277	262	419	250	349	231	215	196	279	201	290	374		
	PrecMin	1111	21	20	58	31	32	16	30	1	25	2	35	27		
	PrecStd	224	61	63	85	55	61	56	46	45	67	53	59	77		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	24.0	27.5	29.1	31.8	34.5	35.1	36.5	34.7	31.3	24.7	22.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	18.2	20.7	25.0	27.5	30.4	33.1	33.1	34.3	32.9	28.4	21.8	19.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	16.1	18.9	22.6	26.1	29.1	32.0	32.1	32.6	31.2	26.4	20.1	17.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	13.8	16.5	20.4	24.2	27.8	30.7	30.9	31.1	29.2	24.3	18.3	15.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.9	12.9	13.8	12.6	12.1	10.7	11.5	12.5	13.6	13.7	10.7		
		MCDBR	13.0	14.6	16.0	15.6	14.7	14.0	12.3	13.8	14.8	15.3	15.5	12.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.326	0.366	0.403	0.454	0.492	0.540	0.540	0.434	0.381	0.342	0.318		
	taud		2.492	2.472	2.383	2.270	2.169	2.099	1.981	1.968	2.258	2.388	2.446	2.516		
	Ebn at Noon		888	922	905	884	838	804	762	753	825	845	851	867		
	Edh at Noon		84	96	114	134	150	161	180	179	128	103	88	78		
(44) All-Sky Solar Radiation	RadAvg		2.45	2.97	4.12	5.35	5.88	6.21	5.90	5.49	4.73	3.85	2.80	2.09		
	RadStd		0.18	0.38	0.39	0.36	0.46	0.47	0.35	0.29	0.43	0.49	0.31	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+147	+106

Nomenclature: See separate page